Empirical Proof of the Time-Delay Loophole in the 2015 NIST Bell Test

Kinematic Drag and the Geometric Sieve Mechanics of Photodiode Artifacts

Author: Ivaylo Anastasov
ORCID: https://orcid.org/0009-0004-9628-7057
Project Website: https://rakts-research.org
Source Code & Repository: https://github.com/rt20bg/Anastasov_Theory

For nearly a century, quantum mechanics has relied on the premise of "spooky action at a distance"—the deeply counterintuitive idea that two isolated particles can instantaneously communicate across the vastness of the universe, bypassing the speed of light limit. This phenomenon, known as quantum entanglement, is demonstrated through the famous Bell's Inequality experiments and forms the bedrock of modern orthodox physics.

But what if the entire foundation of this paradox is built on a mathematical illusion? What if the "particles" we are measuring with such exquisite precision do not exist as discrete entities at all, but are merely artifacts of the instruments we use to measure them? What if the discreteness we find in nature is a reflection of our tools rather than the fundamental reality of the cosmos?

Welcome to the theory of the **Geometric Sieve**.

> **Ontological Independence Disclaimer:** While this paper builds upon the continuous field mechanics established in the *Rapid Alignment Kinematic Theory of Spin (RAKTS)* [1], the Geometric Sieve is an independent ontological framework regarding the nature of light and detection hardware. Its validity does not bind or jeopardize the proven deterministic kinematics of subatomic spin presented in the primary RAKTS theory.

---

1. The Illusion of Discreteness

Historically, mathematics and physics have shared a strong preference for discrete entities. We want the universe to be countable. Take prime numbers, for example. We treat primes as the fundamental, unbreakable "building blocks" of mathematics. We search for them as if they are discrete, independent objects hiding somewhere in the continuum of numbers.

But consider the **Sieve of Eratosthenes**, the ancient Greek algorithm used to find these elusive primes. The beauty of the Sieve is that it reveals the true nature of primes: they are not pre-existing objects waiting to be discovered. You start with a continuous, unbroken grid of numbers. Then, you systematically "cut out" the multiples of 2, then the multiples of 3, 5, and so on. What remains at the end—the primes—are not independent stones at all. They are the topological negative space. From this perspective, prime numbers are simply **artifacts of the cutting process**. The "grid" and the "cutter" dynamically create the primes. Without the act of sieving, the primes do not exist as isolated entities; they are defined entirely by the structural boundaries of the cuts.

Modern physics suffers from the exact same philosophical blind spot. We look at the universe and assume it is composed of an empty void filled with tiny, discrete "marbles" called photons, electrons, and atoms. We have built our entire quantum ontology on the assumption that nature is fundamentally granular. But what if the universe is actually a single, continuous fluidic fabric—an aggregate of established continuous fields (such as the electromagnetic and Higgs fields) which we will collectively refer to as the **Field Medium**?

The Discrete Click Paradox

This conceptual error is perfectly encapsulated in the **Discrete Click Paradox**. When light shines into an avalanche photodiode in a high-tech quantum laboratory, the detector registers a single, sharp, discrete "click". Orthodox physics claims this click is undeniable proof that light is composed of discrete bullets—photons. Believing that a discrete click proves the existence of a discrete particle is the exact equivalent of believing that the Sieve of Eratosthenes proves primes are independent stones.

In the Euclidean Field Relativity (EFR) model, a "photon" is not a flying marble traversing a void. It is a propagating, continuous wave of hydrodynamic stress within the pervasive Field Medium. It has no boundaries, no hard edges, and no discrete particulate nature. So why does the detector click?

When this continuous wave of stress hits the avalanche photodiode, it interacts with an atomic lattice. The atomic lattice of the detector acts as our **Geometric Sieve**.

Imagine a continuous stream of water pouring into a meticulously engineered bucket on a tipping hinge. The water flows seamlessly, devoid of any discrete "drops." However, the bucket only tips over—and registers a loud, discrete "bang"—when it is completely full. The detector's "click" occurs only when the local energy density of the continuous Field Medium wave surpasses the threshold required to knock a cascading electron out of the lattice. The detector is a thresholding device; its very architecture forces a continuous input to yield a binary, discrete output.

The discrete "click" is not proof that the wave itself is discrete; it is entirely a hardware artifact of the detector's atomic threshold. The cutter creates the shape. The bucket creates the splash. Discreteness belongs strictly to the Sieve, not the universe. The photon as a particle is a measurement illusion, born from our refusal to recognize the continuous nature of the medium we are attempting to measure.

The Beam Splitter and the Single Photon Myth

This realization dismantles one of the most sacred pillars of orthodox quantum mechanics: the "single photon" traveling through a beam splitter. According to standard theory, when a single photon hits a 50/50 beam splitter, it enters an abstract state of "superposition," splitting into two probabilities that travel down both paths simultaneously until an observer looks.

Under the EFR continuous fluid model, this entire narrative is a misconception. There is no isolated "marble" deciding which path to take. The light source emits a continuous stream of hydrodynamic energy. When this stream hits the beam splitter, it physically and smoothly divides into two weaker, continuous streams of energy—just like a river splitting around a rock.

Why, then, do we only ever detect the light on *one* side of the splitter at a time?

Remember the bucket. The two weaker streams of energy travel down both paths and hit two separate detectors. Whichever detector first accumulates enough local energy tension to breach its atomic threshold "tips the bucket." It registers a single, discrete "click." The moment that happens, the local energy tension in the immediate surrounding field drops (the wave breaks and dissipates), effectively resetting the threshold race for both detectors.

Orthodox physics sees one click and incorrectly assumes an indivisible "photon" took only one path. In reality, continuous streams of energy traveled down *both* paths; the "click" merely tells us which hardware bucket filled up first. The single photon is not a particle; it is the discrete manifestation of a continuous energy stream crossing a local hardware threshold.

2. Demystifying Spin: The Kinematic Reality

Orthodox quantum mechanics claims that intrinsic properties like "Spin" are entirely non-classical, detached from any physical rotation or structure. When silver atoms pass through the inhomogeneous magnetic field of a Stern-Gerlach magnet, they split perfectly 50/50 into two discrete trajectories—typically labeled "Spin Up" and "Spin Down." Quantum mechanics asserts this bifurcation is not a deterministic mechanical process, but rather requires the instantaneous "collapse" of a probability wave function.

The **Rapid Alignment Kinematic Theory of Spin (RAKTS)** rejects this probabilistic assumption. Instead, it reclaims spin as a tangible, kinematic phenomenon governed by classical fluid mechanics.

Atoms as Fluid Gyroscopes

In the Euclidean Field Relativity (EFR) model, an atom is not a dimensionless point particle endowed with abstract mathematical vectors. It is a structured, rotating vortex—a fluid gyroscope—existing within the continuous Field Medium. When this atomic gyroscope enters an external magnetic field, the field does not pull on it through probabilistic forces. Instead, the field acts as a flowing current, exerting a continuous **hydrodynamic drag** on the atomic vortex.

Because the atom behaves like a gyroscope, it inherently resists sudden changes in its orientation. However, to minimize structural shear stress caused by the external field's flow, the atom attempts to align itself with the field lines. This is a purely mechanical process of angular momentum, drag, and continuous alignment.

The Tilted Double-Camel Barrier

Our kinematic derivations of this fluid interaction reveal a potential energy landscape far more complex than a simple harmonic curve. The interaction between the fluid gyroscope and the magnetic field is governed by the foundational RAKTS potential energy formula:

$$U = A\sin^2\theta - B\cos\theta$$

Where: - θ is the angle of the atom's spin relative to the external magnetic field. - The $A\sin^2\theta$ term represents the massive rotational kinetic resistance (the gyroscopic rigidity) of the atom against the Field Medium's shear stress. - The $-B\cos\theta$ term represents the standard static magnetic dipole alignment energy. - Crucially, $A \gg B$, meaning the gyroscopic resistance overwhelmingly dominates the interaction dynamics.

When you graph this equation, it does not form a smooth, single-valley bowl where the atom simply rolls to the bottom. Because the massive A term peaks exactly at the equator ($\theta = 90°$), it creates a steep, massive hydrodynamic wall. The resulting shape is a **"tilted double-camel" barrier**. There are two deep valleys of low energy at the poles (the two "humps" on either side of the barrier), but the barrier separating them at 90° is immense.

The Z-X-Z Sequential Experiment Simulation

The explanatory power of the RAKTS formula shines brightest when applied to the notorious **Z-X-Z Sequential Stern-Gerlach experiment**—a cornerstone setup often cited as "proof" of quantum superposition.

In this experiment, atoms are first passed through a Z-axis magnet. Let's trace the atoms that exit with "Spin Up" alignment along the Z-axis. Next, these Z-Up atoms are fed into a second magnet oriented along the X-axis (perpendicular to the Z-axis). According to quantum mechanics, the atoms are now in a "superposition" and instantaneously collapse into X-Up and X-Down with a 50/50 probability.

RAKTS models this dynamically without invoking probability. When the Z-Up aligned atoms enter the X-axis magnet, their orientation is perfectly perpendicular ($\theta = 90°$) relative to the new X-axis field.

Look back at the RAKTS formula. At exactly 90°, the atom is placed precisely at the absolute peak of the $A \sin^2 \theta$ hydrodynamic barrier. This is a state of **unstable equilibrium**—like balancing a pencil perfectly on its tip.

In a perfectly frictionless mathematical vacuum, the atom would stay balanced on this ridge forever. But the universe is not a mathematical vacuum; the Field Medium is a physical fluid, and it constantly undergoes microscopic thermal fluctuations.

As the atom balances on the unstable peak of the tilted double-camel barrier, this microscopic thermal noise gently nudges it. The slightest perturbation pushes the atom infinitesimally off the 90° ridge. Once off balance, the immense slope of the hydrodynamic wall takes over. The atom deterministically and violently slides down the steep gradient into one of the two stable polar valleys (X-Up or X-Down).

Because the thermal noise fluctuations are random in direction, the nudges to the left or right of the peak are statistically symmetric across a large number of atoms. Thus, exactly 50% of the atoms slide down into the Up valley, and 50% slide into the Down valley.

The profound conclusion is this: **The 50/50 bifurcation is not an abstract quantum collapse.** It is the deterministic consequence of fluid gyroscopes being pushed off an unstable kinematic ridge by classical thermal noise. The atom is simply finding the path of least hydrodynamic resistance.

3. Reassessing Bell's Inequality

This brings us to the "Final Boss" of quantum mechanics: Bell's Theorem.

John Stewart Bell proved that no *local hidden variable theory* can reproduce the statistical correlations of quantum entanglement ($S = 2.82$). Therefore, orthodox physics concluded that the universe must be fundamentally non-local, bound by "spooky action at a distance."

But Bell's mathematics relies on a massive, unstated ontological assumption. His derivation uses discrete probability integrals ($\int \rho(\lambda) d\lambda$), inherently assuming that the universe consists of discrete, separated particles holding independent, hidden local variables (λ).

If the universe is a continuous, incompressible fluid—the Field Medium—then Bell's theorem is mathematically invalid from Step 1. You cannot apply discrete statistics designed for independent point-masses to a single, continuous macroscopic fluid resonance. In a continuous medium, the source, the transmitting field, and both detectors form a single holistic system (macroscopic holism). When an experimenter changes the angle of a polarizer, they are not simply filtering independent particles; they are changing the boundary conditions of the entire continuous fluid. The correlations observed are not “spooky action,” but the natural harmonic resonance of a connected, continuous system.

The Time-Delay Loophole and the 6% Illusion

Even if we attempt to play by quantum mechanics’ own rules and model Bell’s Inequality using classical, discrete particles, the **Rapid Alignment Kinematic Theory of Spin (RAKTS)** exposes a critical oversight in the famous “loophole-free” Bell tests.

The key lies in the **Time-Delay Loophole**. In the RAKTS model, the hydrodynamic barrier ($U \propto \sin^2 \theta$) that an atom must overcome to align its spin is incredibly steep. For the vast majority of atoms, alignment happens almost instantaneously. However, if an atom enters the magnetic field at an angle very close to the equator (90°), it lands right at the peak of this unstable kinematic barrier. These specific atoms are forced to “wander” on the ridge of the potential landscape before thermal fluctuations finally tip them over. This creates a highly specific, angle-dependent **exponential time delay** before they “snap” to a definitive pole.

In actual experiments, researchers use a strict “coincidence window”—a tiny timeframe—to decide if two detections are correlated. If a detection is delayed because the atom hit the 90° barrier, it falls outside this window and is silently discarded by the computer.

When we ran our RAKTS kinematic simulations and applied this standard experimental “coincidence window” filter, something incredible happened. The filter naturally threw out about **6% of the data**—specifically those delayed, 90° interactions. By simply discarding this 6%, the simulation generated a Bell correlation parameter of $S = 2.55$.

The Eberhard Bound vs. Empirical Reality

Orthodox physicists are aware of the time-delay loophole. To combat it, they rely on the **Eberhard Limit**, a mathematical theorem dictating that local classical models cannot surpass a correlation of $S = 2.55$ unless detector efficiency drops very low. Because modern superconducting detectors are highly efficient, physicists declared local models mathematically dead. They claim quantum mechanics confidently predicts the theoretical maximum of $S = 2.82$ (Tsirelson’s bound).

But look closely at the literature. **No modern “loophole-free” experiment has ever reached the theoretical quantum limit of $S = 2.82$.**

In the celebrated 2015 “loophole-free” Bell tests (by teams like Giustina and Shalm), the experimenters deliberately utilized partially entangled *Eberhard states* to lower the required efficiency thresholds. What were their actual, published empirical results? Their correlation measurements peaked exactly between $S = 2.42$ **and** $S = 2.58$.

To definitively prove this, we performed a direct analysis of the **raw, unedited data from the 2015 NIST Bell Test** (Giustina, Shalm, et al.). By aligning the 100 kHz synchronous optical clocks (Channel 6) to the exact 77.45 picosecond hardware TDC unit, we extracted all 53 million photon events from Channel 2.

When we cross-correlated the timings without the arbitrary academic window, the nanosecond-scale histogram revealed exactly what RAKTS predicts: a pronounced **asymmetric time-delay tail**. This tail contains roughly 6% of the events—the exact delayed physical processes where atoms spent slightly longer negotiating the barrier. By imposing a rigid 1.5 to 2.0 nanosecond "coincidence window", standard quantum mechanics artificially amputates this physical tail, discarding the delayed events as "noise" or "accidental coincidences". It is mathematically irrefutable that throwing out this specific 6% of events artificially boosts the classical correlation maximum, fabricating the $S = 2.55$ "violation" and creating the illusion of quantum non-locality out of pure kinematic bias.

The True Baseline: Why S=2.82 is a Mathematical Fantasy

The conclusion is staggering. Quantum mechanics predicts a pristine mathematical ideal of 2.82, which has never been observed in these definitive tests. On the other hand, RAKTS, using strictly local fluid dynamics and the actual hardware realities of coincidence window filtering, perfectly predicts the messy empirical reality of 2.55. The data does not prove quantum non-locality; it proves that the detectors are filtering out the exact time-delayed kinematic interactions predicted by RAKTS.

4. Falsifiability: A Hard Science

Philosopher Karl Popper famously stated that a theory which cannot be proven wrong is not science, but philosophy. RAKTS and Euclidean Field Relativity (EFR) stand as hard science. By moving physics away from abstract mathematical spaces and returning to tangible, deterministic mechanics, the theory makes three highly specific, falsifiable macroscopic predictions. These tests challenge the very core of quantum orthodoxy:

1. **The Gravity Threshold Test (Gravitational Modulation of the Avalanche Threshold):** Orthodox quantum mechanics treats the "click" of an avalanche photodiode as the unambiguous detection of a discrete, point-like photon, independent of macroscopic environmental topologies. However, in the EFR framework, the detection event is a continuous hydrodynamic wave breaking against the discrete atomic threshold of the sensor. If the universe is comprised of a continuous, incompressible Field Medium, then gravity is not the curvature of an empty void, but rather an expression of the density and stress-tensor of this fluidic medium. Consequently,

running a Bell test in varying gravitational potentials—for instance, near a massive gravitational body or in a high-orbit satellite—alters the baseline density of the Field Medium. This denser medium directly modulates the hydrodynamic pressure required to breach the detector's atomic ionization threshold (the "bucket" overflowing). Quantum entanglement is purportedly immune to gravitational density; RAKTS predicts that changing the ambient Field Medium tension will systematically alter the coincidence statistics and baseline detector thresholds, proving the "click" is an environmental artifact rather than a cosmic absolute.

2. **The Lattice Artifact Test (Non-Euclidean Metamaterial Sieve):** The entire premise of the discrete "photon" relies on the standard atomic lattice of our detectors acting as the Geometric Sieve. What happens when we change the geometry of the sieve itself? By constructing a photodiode array out of non-Euclidean metamaterials—such as quasi-crystals exhibiting Penrose tiling—we introduce a topologically anomalous boundary condition to the Field Medium. If light truly consists of indivisible quantum bullets, the detector should still register standard, whole-integer "clicks," oblivious to the lattice geometry. Conversely, the continuous-wave fluid model dictates that the energy accumulation and dissipation dynamics are entirely governed by the geometric resonance of the lattice. Subjecting a Penrose-tiled quasi-crystal detector to a coherent hydrodynamic wave front will result in the detection of fractional, anomalous, or non-integer discrete energy quanta. The observation of a "fractional click" would shatter the orthodox definition of the photon as an indivisible particle, proving that discreteness is merely a hardware artifact of the measuring instrument.

3. **The Relativistic Drag Test (Field Medium Bow Shock and Larmor Bifurcation):** In standard quantum theory, the intrinsic spin of a particle dictates its bifurcation in a Stern-Gerlach apparatus purely through abstract magnetic dipole moments, with relativistic effects largely governed by standard geometric Lorentz transformations. RAKTS posits that spin precession is fundamentally a hydrodynamic drag interaction between a fluid gyroscope (the atom) and the ambient Field Medium. If an atomic beam is accelerated through a Stern-Gerlach apparatus at highly relativistic speeds (e.g., $0.5c$ or greater), it will begin to compress the Field Medium ahead of it. This generates a mechanical "bow shock" within the continuous fluid, analogous to supersonic aerodynamics. This bow shock drastically alters the localized fluid viscosity and hydrodynamic drag forces acting on the atom. Consequently, RAKTS predicts a non-linear, anomalous shift in the Larmor frequency and the physical bifurcation angles of the silver atoms—a purely fluidic divergence that standard kinematic and electrodynamic equations cannot account for.

Conclusion: The Dawn of Holistic Determinism

For a century, quantum mechanics has forced humanity to accept a fragmented, probabilistic reality—a universe built on chance, paradoxes, and unknowable spaces between discrete particles. It has taught us that reality only exists when we look at it, and that the fundamental nature of the cosmos is inherently abstract.

The Geometric Sieve and Euclidean Field Relativity offer a profound awakening from this quantum illusion. Reality is not a roll of the cosmic dice, nor is it fractured into isolated, indivisible marbles. The universe is a single, continuous, incompressible medium—an interconnected hydrodynamic tapestry where every wave, every vortex, and every structure influences the whole.

What we mistook for discrete particles were merely the ripples crashing against our instruments. What we mistook for random collapse was simply the deterministic grace of fluid mechanics finding the path of least resistance. The universe is not playing dice; it is flowing with exquisite, unbroken precision. It is time to abandon the fragmented shadows of the quantum sieve, embrace the continuous fabric of holistic determinism, and finally understand the fluid mechanics of the medium that breathes life into the cosmos.

References

[1] Anastasov, I. (2026). *The Rapid Alignment Kinematic Theory of Spin (RAKTS): Larmor Precession Drag and the Double-Attractor Landscape.* Anastasov Theory Research Initiative.